Resting delicately inside the barrel, a Styrofoam dart waited patiently for its share of kinetic energy. The dart gun, already brimming with potential energy generated by the pumping of Ms. Pino's arms, waited patiently for a signal to fire. And I waited patiently as a first-year Physics AP student to learn about projectile motion and the laws of physics galore. The air grew thick with anticipation as trouble-free molecules of air incessantly collided with each other, and I opened my eyes wider. When she finally pulled the trigger, my eyes raced from one side of my face to the other as I tried to chase the path of the dart from gun to window.

"Projectile motion," said my teacher, "is the roughly parabolic path traced by a free-falling object. In this case, the dart is a projectile." Making a mental note, I tried to connect concept with formula, projectile motion with its counterpart, $x = vt + \frac{1}{2} at^2$. At first, I refused to believe that pure equations could translate to the physical world, but when I finally learned to connect the two, I eagerly allowed myself to absorb and apply the new information.

As I processed formulas, I began to notice the world around me – from a physics standpoint, that is. Playing tennis, I saw projectile motion every time the ball flew. When I went to sip water from the drinking fountain, I saw projectile motion in the parabolic curve traced by the path of water. With each new subject brought up by Ms. Pino, I gained a new perspective on the world. Appreciating what I now understood, I tried to put the information to good use by conducting experiments to confirm the Coriolis effect, to attempt to finalize the Grand Unified Theory of Everything (GUT, of everything), or to disprove Newton's gravity by embracing Einstein's space-fabric. It was here that I first became acquainted with the field of engineering.

Ms. Pino had asked me if I would like to take part in a bridge-building contest. Run by the naval academy West Point, the contest required the participant to design and test a working bridge. The winning design would be judged on how expensive the materials would cost, a direct measure of efficiency.

Eager to start, I downloaded the bridge-design software that night and jumped blindly into the program. For one hour I twiddled with joints and bars, hoping to come across a bridge by chance – but the odds were against me because, in the end, I remember staring at my sad creation, a clutter of tubes and bars that refused to even stand up on its own. But far from feeling disheartened, I felt instead motivated, intrigued, and challenged by the intellectual demand placed upon my mind. Typing "truss bridges" into the Yahoo search engine, I found a helpful file explaining the basics behind bridges. Later, using two manila folders, glue, scissors, a chunk